

RFC CRITICAL BUG & AMENDMENT REPORT

Verified technical defects and amendment recommendations based on the supplied RFC index and official RFC Editor errata records.

SCOPE The supplied attachment is an RFC index/catalogue, not the full text of every RFC.	METHOD Cross-check RFCs represented in the index against official RFC Editor errata and prioritize verified technical defects.	INTEGRITY RULE No speculative vulnerability is presented as a confirmed bug. Severity below is an engineering assessment.
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1. Executive Findings

The review identified several verified technical errata in foundational RFCs represented in the supplied corpus. The most consequential issues concern IPv4/IPv6 fragmentation, TLS 1.3 handshake and cryptographic encoding rules, and HTTP grammar. These are real specification defects already recognized by the RFC Editor, not newly invented claims.

Priority	RFC	Verified defect	Engineering impact	Amendment disposition
HIGH	791	IPv4 reassembly length calculation can use the wrong IHL when fragments differ.	Incorrect reassembly accounting and interoperability risk.	Correct the reassembly algorithm; add divergent-IHL tests.
HIGH	8200	IPv6 Fragment Offset was originally defined relative to the wrong packet portion.	Can cause inconsistent fragment positioning/reassembly.	Retain verified correction in the normative IPv6 text.
HIGH	8446	CertificateRequest rejection condition was too broad.	Clients can reject a valid main-handshake CertificateRequest.	Constrain the condition to post-handshake/application-traffic context.
HIGH	8446	ECDSA signature encoding wording was ambiguous.	Potential incompatible signature serialization/parsing.	Specify the TLS-compatible DER ECDSA structure explicitly.
HIGH	8446	PSK binder text overstates transcript binding for external PSKs.	Can misstate the security property of external-PSK handshakes.	Qualify the statement for NewSessionTicket-derived PSKs.
MED-HIGH	9110	Range ABNF omitted optional whitespace permitted by examples.	Strict grammar-driven parsers may reject valid forms.	Add OWS to the grammar and conformance tests.

2.1 RFC 791 — IPv4 Fragment Reassembly

Status: RFC Editor EID 6356; verified technical

Defect: Section 3.2's reassembly procedure used the IHL of the fragment being processed in a total-length calculation. The verified correction specifies use of the first fragment's IHL because fragment headers can differ.

Impact: An implementation following the faulty procedure can calculate an incorrect reassembled length when options cause different IHL values. This is a protocol correctness and interoperability defect.

Proposed amendment: Replace the algorithmic reference with the first fragment's IHL and state the rule normatively. Add tests with different IHL values across fragments.

Evidence: <https://www.rfc-editor.org/errata/eid6356>

2.2 RFC 8200 — IPv6 Fragment Offset

Status: RFC Editor EID 5945; verified technical

Defect: The original Fragment Header description used the wrong conceptual reference point for Fragment Offset. The verified correction resolves this in terms of the Extension & Upper-Layer Headers part.

Impact: Fragment offset interpretation directly affects reassembly. Different interpretations can place payload at different offsets and break interoperability.

Proposed amendment: Keep the verified correction in the normative text and align all terminology around unfragmentable and fragmentable parts.

Evidence: <https://www.rfc-editor.org/errata/rfc8200>

2.3 RFC 8446 — Post-Handshake CertificateRequest

Status: RFC Editor EID 8795; verified technical

Defect: The original rule was broad enough that a client could treat a main-handshake CertificateRequest as an unexpected message when it had not advertised post-handshake authentication.

Impact: This can cause a conforming client to abort a valid main handshake, producing a direct interoperability failure.

Proposed amendment: Limit the fatal-alert condition to a CertificateRequest encrypted under application traffic keys in the post-handshake case. The erratum notes this correction is addressed in 8446bis.

Evidence: <https://www.rfc-editor.org/errata/eid8795>

2.4 RFC 8446 — ECDSA Encoding Ambiguity

Status: RFC Editor EID 5868; verified technical

Defect: The ECDSA description did not sufficiently disambiguate the TLS signature structure from a conflicting ASN.1 definition in another authoritative source.

Impact: Cryptographic serialization must be unambiguous. Different interpretations can cause valid TLS signatures to be rejected or parsed inconsistently.

Proposed amendment: Explicitly reference the TLS-compatible DER ECDSA-Sig-Value structure and align the normative references.

Evidence: <https://www.rfc-editor.org/errata/eid5868>

2.5 RFC 8446 — External-PSK Binder Description

Status: RFC Editor EID 8803; verified technical

Defect: The PSK binder explanation generalized transcript binding in a way that does not hold for externally provisioned PSKs in the same manner as NewSessionTicket-derived PSKs.

Impact: This is a security-property specification error: it can lead reviewers or implementers to infer stronger provenance/binding guarantees than the protocol provides.

Proposed amendment: Qualify the description so transcript binding through the original handshake is attributed only where the PSK was established through a NewSessionTicket.

Evidence: <https://www.rfc-editor.org/errata/eid8803>

2.6 RFC 9110 — Range ABNF

Status: RFC Editor EID 7306; verified technical

Defect: The Range grammar omitted optional whitespace after the range-unit delimiter even though examples allow that form.

Impact: Implementations generated directly from the ABNF can reject input that the document examples imply is valid, creating parser interoperability problems.

Proposed amendment: Add optional OWS to the ABNF and add grammar/example conformance vectors.

Evidence: <https://www.rfc-editor.org/errata/eid7306>

3. RFC 793 — Historical TCP Defects

RFC 793 is obsolete and has been superseded by RFC 9293. The RFC Editor records 11 verified technical errata for RFC 793, including corrections/clarifications affecting the TCP state diagram, simultaneous open, retransmission timeout, event processing, sequence-number accounting, and reset behavior. The correct amendment strategy is therefore to verify that the current TCP specification incorporates the relevant corrections rather than creating a new amendment to RFC 793.

Evidence: RFC Editor RFC 793 errata and EID 573. <https://www.rfc-editor.org/errata/rfc793>

4. Cross-RFC Critical Risk Themes

Fragmentation: packet metadata and offset calculations must be tested with non-default headers/options.

Grammar: ABNF must be tested against every normative example and parser implementation.

Cryptography: signature structures need one unambiguous serialization definition.

Security properties: protocol text must distinguish resumption PSKs from external PSKs.

State machines: diagrams, prose, and event-processing rules must be tested as one executable model.

5. Amendment/Erratum Record Template

Field	Required content
RFC / section	Exact RFC and affected section.
Existing erratum	Official erratum ID/status, if applicable.
Problem	Reproducible normative defect.
Correction	Exact replacement rule/grammar.
Impact	Interoperability, security, parsing, or state-machine consequences.
Compatibility	Expected effect on deployed implementations.
Test vector	Packet/message/handshake sequence proving the issue.
Disposition	Erratum, update RFC, bis revision, or obsolescence.

6. Limitations and Next Phase

The attached source is an RFC index, not the full body of every RFC. Consequently, this report is a verified-errata-based critical defect report, not a claim that every RFC in the index has been exhaustively audited for undiscovered vulnerabilities. The RFC Editor states that published RFCs do not change and that errata are used to record errors; verified errata are reviewed by the appropriate parties.

A full corpus audit should retrieve the complete RFC texts, correlate verified/reported/held/rejected errata, inspect normative grammar and state machines, and generate executable interoperability tests. Obsolete RFCs should be mapped forward to their current replacement standards.

Official source: RFC Editor Errata — <https://www.rfc-editor.org/errata.php>

Official source: RFC Editor RFC Index — <https://www.rfc-editor.org/rfc-index/>

Prepared: 26 August 2026. Severity labels are engineering assessments; erratum status is the official RFC Editor status.